

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
 School of Engineering and Engineering Technology
 Department of Electrical and Electronics Engineering
 B. Eng. Degree Examination, Second Semester 2023/2024 Session
 ICT 304: Electromagnetic Field Theory; Date: 27th September, 2024:
 3 HOURS.



General Instruction: Answer ALL QUESTIONS in Section A and ONE QUESTION EACH from Section B, C and D.

Parameters: $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

SECTION A

1. Which of the following is the correct expression for the dot product of two vectors $\mathbf{A} \cdot \mathbf{B}$
 (A) $|\mathbf{A}||\mathbf{B}| \sin \theta$ (B) $|\mathbf{A}||\mathbf{B}| \cos \theta$ (C) $\frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|}$ (D) None of the above.
2. In cylindrical coordinates, the differential length $d\mathbf{l}$ is expressed as: (A) $dr d\theta dz$ (B) $\rho d\rho d\theta dz$ (C) $\rho d\rho dz d\theta$ (D) None of the above.
3. The electric field intensity at a point due to a point charge is given by which law? (A) Gauss's Law (B) Ampère's Law (C) Coulomb's Law (D) Faraday's Law
4. Which of the following is the correct expression for the electric potential V at a distance r from a point charge Q ? (A) $V = \frac{Q}{4\pi\epsilon_0 r^2}$ (B) $V = \frac{Q}{4\pi\epsilon_0 r}$ (C) $V = \frac{Q}{r}$ (D) $V = Q$
5. The electric field due to an infinite line charge is proportional to: (A) $\frac{1}{r^2}$ (B) $\frac{1}{r}$ (C) $\frac{1}{r^3}$ (D) Constant.
6. Which of the following is a direct result of Gauss's Law? (A) $\nabla \times \mathbf{E} = 0$ (B) $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$ (C) $\nabla \times \mathbf{H} = \mathbf{J}$ (D) $\nabla \cdot \mathbf{B} = 0$.
7. What is the electric field intensity at the center of a uniformly charged spherical shell?
 (A) Zero (B) $\frac{Q}{4\pi\epsilon_0 r^2}$ (C) $\frac{Q}{\epsilon_0 r^2}$ (D) None of the above
8. The Laplacian of a scalar function is denoted as: (A) $\nabla \times \mathbf{A}$ (B) $\nabla^2 V$ (C) $\nabla \cdot \mathbf{A}$ (D) ∇V
9. The magnetic field intensity $\mathbf{H}$ is related to the current density $\mathbf{J}$ by: (A) Coulomb's Law (B) Ampère's Circuital Law (C) Gauss's Law for Magnetism (D) Faraday's Law
10. The Stokes theorem is used to relate: (A) Surface integrals to volume integrals (B) Line integrals to surface integrals (C) Surface integrals to line integrals (D) Volume integrals to line integrals
11. The vector field representing the magnetic field is denoted by the symbol _____.
12. The electric potential difference between two points is defined as the work done per unit _____.
13. In Cartesian coordinates, the differential surface area $d\mathbf{A}$ for a rectangle with its normal in the $+\mathbf{a}_x$ is _____.
14. The divergence of a vector field gives information about the net _____ leaving a point.
15. A unit vector has a magnitude of _____.
16. The electric flux through a surface is the surface integral of the _____.
17. The line integral of the electric field around a closed path is zero. This is a direct result of _____'s second law.

18. A magnetic dipole is created by a closed loop of _____.
19. The SI unit of the electric field intensity is _____.
20. For a point charge, the electric field intensity decreases as the square of the _____.
21. The electrostatics, charges are always in constant motion (T/F).
22. The electric field intensity of an infinite sheet of charge decrease with distance distance (T/F).
23. Coulomb's law is used for symmetrical shapes while Gauss law is used for general shapes in electrostatics. (T/F).
24. Magnetic monopoles exist. (T/F).
25. In Magnetostatics, current is time-varying (T/F).

SECTION B

1. (a). What do you understand by the following terms in electromagnetic field theory :
 (i). Gaussian surface. (3 Marks)
 (ii). Amperian circuit path. (3 Marks)
- (b). Given two vectors $C = 6a_x - 2a_y + a_z$ and $D = 3a_x + a_y - 4a_z$
 (i). Find the angle between vectors C and D using the dot product. (5 Marks)
 (ii). Find the vector perpendicular to both C and D using the cross product. (6 Marks)
- (c). (i). State the Biot-Savart law and provide its mathematical representation. (3 Marks)
 (ii). An infinite sheet of charge with uniform surface density $-2\rho_s$ is located at $z = 0$ (x-y plane), and another infinite sheet with density $5\rho_s$ is located at $z = -3m$, both in free space. Determine E everywhere. (5 Marks)
2. (a). (i). Given two points in a cartesian coordinate system, $P_1 (2,3,4)$ and $P_2 (-1,2,-3)$, show the spatial diagram and calculate the distance between the two point. (3 Marks)
 (ii). Convert the Cartesian coordinates $(3, 4, 5)$ to cylindrical coordinates. (3 Marks)
- (b). (i). For a vector field $F = ya_x + xa_y + z^2a_z$, compute the curl of F at point $(1, 2, 1)$. What does your answer implies. (4 Marks)
 (ii). Given that the scalar field $\phi = 4xa_x + 3y^2a_y - 2za_z$, compute the gradient at the point $(1, -1, 2)$. What does your answer implies. (4 Marks)
- (c). (i). State the ampere's law and use it to derive the Maxwell's first equation of magnetostatics in both integral and differential form. (6 Marks) (3)
 (ii). Two point charges $q_1 = 5\mu C$ and $q_2 = -2\mu C$ are located at points $(0, 0, 0)$ and $(0, 0, 4)$ meters respectively. Calculate the electric potential V at point $(0, 0, 2)$ due to these two charges. (5 Marks)

SECTION C

3. (a). Four charges of $10 mC$ each are located in free space at points with Cartesian coordinates $(-3,0,0)$, $(3,0,0)$, $(0,-3,0)$, and $(0,3,0)$.
 (i). Find the force on a $20-mC$ charge located at $(0,0,4)$. All distances are in meters. (4 Marks)

- (ii). Calculate the Electric field intensity at the point (0, 0, 4). (3 Marks)
- (b). (i). Calculate the circulation of the vector field $\mathbf{F} = (y-x)\mathbf{a}_x + (x+y)\mathbf{a}_y$ around the square vertices at (0, 0), (1, 0), (1, 1), (0, 1) in the clockwise direction. (5 Marks)
- (ii). Let $\mathbf{A} = y\mathbf{a}_x + z\mathbf{a}_y + x\mathbf{a}_z$. Find the flux of $\mathbf{A}$ through surface $y = 1$, $0 < x < 1$ and $0 < z < 2$. (4 Marks)
- (c). Given a magnetic field intensity is given as $\mathbf{H} = 2xy\mathbf{a}_x + 3x^2z\mathbf{a}_y + 4xyz\mathbf{a}_z$ where x, y, z are in meters and the charge density ρ is given as $5(x+y+z)$ in C/m^3 .
- (i). Using Maxwell's first equation of magnetostatics in the differential form, derive the expression for the current density $\mathbf{J}$ as a function of the charge density ρ . (4 Marks)
- (ii). Calculate the total current passing through a surface at $x = 1$ in the region bounded by $0 \leq y \leq 2$ and $0 \leq z \leq 3$. (5 Marks)
4. (a). (i). Evaluate the divergence of this vector field: $\mathbf{B} = \rho z^2\mathbf{a}_\rho + \rho \sin^2\theta\mathbf{a}_\theta + 2\rho z \sin^2\theta\mathbf{a}_z$. (3 Marks) ✓ *Divergence*
- (ii). Given that $\mathbf{D} = 2\rho z^2\mathbf{a}_\rho + \rho \cos^2\theta\mathbf{a}_\theta$, evaluate $\int_V \nabla \cdot \mathbf{D} dv$ over the region defined by $2 \leq \rho \leq 5$, $-1 \leq z \leq 1$, $0 \leq \theta \leq 2\pi$. (5 Marks) ✓ *Divide by $\rho^2 \sin\theta$*
- (b). (i). Determine $\mathbf{D}$ at (4, 0, 3) if there is a point charge at $-5\pi\text{mC}$ at (4, 0, 0) and a line charge $3\pi\text{mC/m}$ along the y axis. (4 Marks) ✓
- (ii). State the Maxwell's second law of magnetostatics in the integral form and derive the differential form of it. Describe briefly their physical meaning. (6 Marks) ✓
- (c). (i). Determine if this Electric field is irrotational and/or divergenceless:
 $\mathbf{E} = -\frac{V_0}{d}\mathbf{a}_x$ (4 Marks)
- (ii). Find the Laplacian of the following scalar fields and compute the value at the specified point. $U = x^3y^2e^{yz}$ at (1, -1, 1). (3 Marks) ✓

SECTION D

5. (a). (i). With illustrations and equations, state the 4 categories of vector fields. (4 Marks)
- (ii). Evaluate both sides of the divergence theorem for the vector field
 $\mathbf{H} = 2xy\mathbf{a}_x + (x^2 + y^2)\mathbf{a}_y + 2yz\mathbf{a}_z$ (5 Marks)
- (b). (i). Point charge $Q_1 = 300\mu\text{C}$, located at (1, -1, -3)m experience a force $\mathbf{F}_1 = 8\mathbf{a}_x - 8\mathbf{a}_y + 4\mathbf{a}_z$ N due to point charge Q_2 at (3, -3, -2)m, determine Q_2 . (4 Marks)
- (ii). Use the Gauss' law and divergence theorem to derive the first Maxwell's equation of electrostatics. What is the physical meaning of both equations. (5 Marks)
- (c). (i). Line $x = 0, y = 0, 0 \leq z \leq 10\text{m}$ carries current 2 A along $\mathbf{a}_z$. Calculate $\mathbf{H}$ at points (5, 0, 0). (4 Marks)
- (ii). In a tabular form, differentiate between Electrostatics and Magnetostatics in the following attributes as shown in the table below: (3 Marks)

Attributes	Electrostatics	Magnetostatics
Sources		
Field and Fluxes		
Constitutive Parameters		
Governing Maxwell's Equations (Integral form)		
Governing Maxwell's Equations (Differential form)		
Force on charge q		

6. (a). Given that $\mathbf{F} = x^2 y \mathbf{a}_x + y \mathbf{a}_y$, Find

(i). $\oint_L \mathbf{F} \cdot d\mathbf{l}$ where L is shown in Figure 1. (4 Marks) ✓

(ii). $\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ where S is the area bounded by L in Figure 1. (4 Marks) ✓

(iii). Is Stoke's theorem satisfied? (1 Marks) ✓

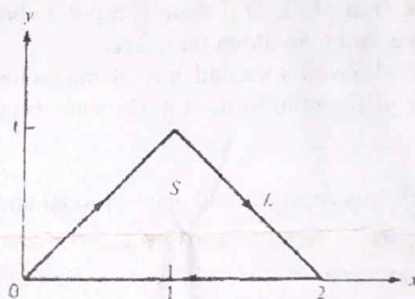


Figure 1

(b). (i). Derive Maxwell's second equation for electrostatics in the integral form and differential form. What does both equations tell us. (5 Marks) ✓ ③

(ii). Three point charges $Q_1 = 1 \text{ mC}$, $Q_2 = -2 \text{ mC}$, and $Q_3 = 3 \text{ mC}$ are, respectively, located at $(0, 0, 4)$, $(-2, 5, 1)$, and $(3, -4, 6)$. Find the potential V at $P(-1, 1, 2)$. (4 Marks)

(c). (i). What are the fundamental differences between electric and magnetic fields? (3 Marks)

(ii). Determine if this Magnetic field is irrotational and/or divergenceless:

$$\mathbf{B} = \frac{\mu_0 I}{2\pi d} \mathbf{a}_\phi$$

(4 Marks)